Symbolic circuit reference: selected identities and comparisons

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This appendix records exact identities and scoped comparisons. The revised manuscript and its current independent compiler table are authoritative. The atlas is not claimed to dominate every tool or publication.

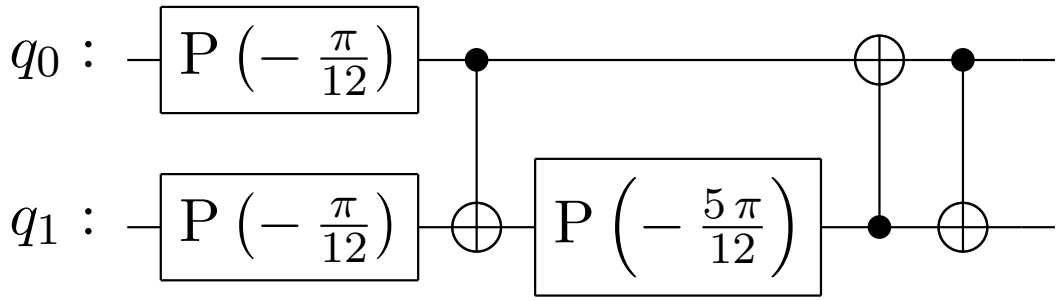
1 Controlled fermionic SWAP

The literal CX/U3 circuit has four CNOTs and six locals. The interaction count is optimal among arbitrary Ising-class interactions without ancillas; the local count is an upper bound. BQSKit independently finds four-CNOT, six-local circuits, so discovering the upper bound is not a new compiler advantage. The lower theorem extends to all controlled balanced involutions. The three-qubit hardware family of Warren et al. is related physical context, not a proof of these CNOT counts [1].

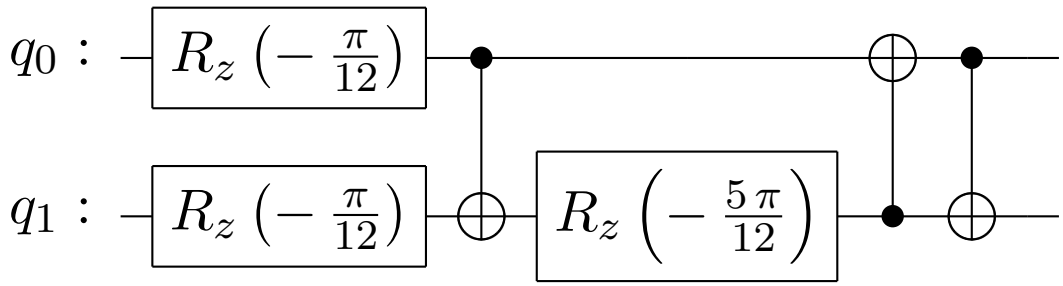
2 Sycamore phase contract

For SYC = fSim($\pi/2, \pi/6$), the exact P/CX circuit and the projective Rz/CX circuit below each use three entanglers and three locals. In the Rz circuit, SYC = $e^{-7i\pi/24}C$. The phase-polynomial certificate proves the three-phase lower bound only in the CNOT/P subalphabet; it does not establish an unrestricted local minimum. The former tomography-figure comparison has been withdrawn.

cx_h_phase



ibm_cx



3 Current independent fixed-target comparison

The table gives E/L in CX/U3 and the best verified recorded route for each tool. Inputs are independent library words or target matrices, not the atlas output. A dash excludes all recorded outputs on that route that fail its replay threshold or budget. Quartz and QUESO optimize total NAM gates; their locals are then fused to U3 without changing the entangling skeleton.

Gate	Qiskit	tket	BQSKit	Quartz	QUESO	Exact atlas
SYC	3/8	3/7	3/4	4/5	4/5	3/3
iSWAP	2/3	2/4	2/4	2/3	2/3	2/3
SSWAP	3/8	3/7	3/5	–	3/8	3/7
fFredkin	5/13	10/15	4/6	9/10	13/10	4/6
CCX	6/8	6/8	6/8	6/8	6/8	6/8
CSWAP	7/10	7/8	7/9	8/8	8/8	7/10
C3X	14/19	14/16	14/16	14/16	14/16	14/17
C4X	36/41	36/41	–	30/33	36/36	36/41

BQSKit matches fFredkin and beats the SSWAP local count. tket and BQSKit beat the C3X local count. Quartz's 30-CX C4X circuit beats the current 36-CX atlas entry. These losses are retained. The earlier eleven SDK entangler wins were all fFredkin in different alphabets.

4 Historical broad local-count audit

The CSV files retain comparisons against the older, explicitly identified SDK routes, including source-fed routes. They are not comparisons against all compiler strategies. Counts and parameter-description score are different objectives; a smaller L alone does not prove a better archived score.

Historical status	Cells
equal_E_equal_locals	2108
equal_E_fewer_locals	553
equal_E_more_locals	1
fewer_entanglers	11

5 Established native bounds and proof scope

The universal two-B construction [2] and the SQiSW two-pulse Weyl region and three-pulse universal construction [3] are prior results. The released exact charts implement them with explicit scalar handling and branch definitions. The new invariant iff proof rewrites the known SQiSW region;

it is not a new universal pulse cap. The 192 one-qubit phase-resolved Clifford circuits are a finite reference, not a large unexplored class.

6 Second-review extensions and resource gaps

The theory paper now gives normal forms for every spectral stratum of two-qubit involutions. Trace-± 2 reflections interpolate between CCZ and Fredkin, with proved interior bounds of four to eight CNOTs and three to six unrestricted interactions. These are partial bounds, not an exact interior classification. The SQiSW result is presented as a polynomial reformulation of the known reachable region for symbolic-family reasoning.

The native library now has an exact two-SYC controlled-phase chart and a literal one-TK2 DCX circuit. Remaining SYC and square-root-SWAP gaps are annotated in the atlas against separate numerical witnesses; a family sample is not a uniform bound or an exact symbolic word. Remaining TK2 gaps are flagged against its known Cartan capacity. The standalone supplement contains the gap ledger and complete empirical evaluation.

7 Reproduction

The companion JSON and CSVs retain archive hashes, exact identities, older comparison settings, and every local-count regression. The revised review bundle additionally includes the stronger SDK/Quartz/QUESO outputs, failures, optimality ledger, theorem checks, and both complete atlas PDFs and sources. UniversalQCompiler has no executed comparison because its licensed kernel was unavailable. No numerical sample proves a symbolic family identity or an unrestricted global minimum.

References

[1] C. W. Warren, J. Fernández-Pendás, S. Ahmed, et al., Extensive characterization and implementation of a family of three-qubit gates at the coherence limit, npj Quantum Information 9, 44 (2023), doi:10.1038/s41534-023-00711-x.

[2] J. Zhang, J. Vala, S. Sastry, and K. B. Whaley, Minimum construction of two-qubit quantum operations, Physical Review Letters 93, 020502 (2004), arXiv:quant-ph/0312193.

[3] C. Huang et al., Quantum instruction set design for performance, Physical Review Letters 130, 070601 (2023), arXiv:2105.06074.